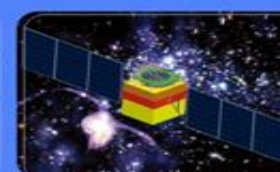


R&D of High Light Yield and High Density Glass Scintillator for High Energy Physics



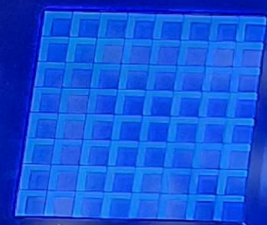
WWW.IHEP.CAS.CN



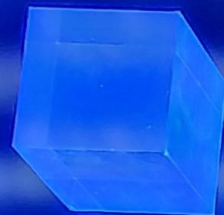
Sen QIAN

qians@ihep.ac.cn; On Behalf of the GS R&D Collaboration Group

The Institute of High Energy Physics, CAS



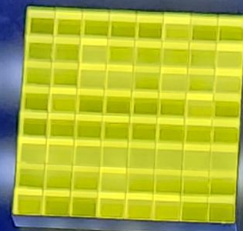
LYSO



LYSO



LYSO



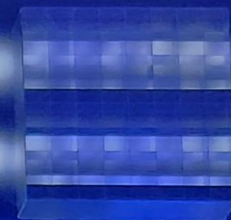
GAGG



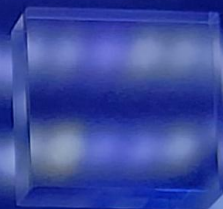
GAGG



GAGG



BGO



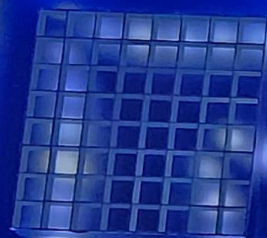
BGO



BSO



PWO



PbGlass



Quartz



NaI



CLB



CLTC



BaF₂



PS



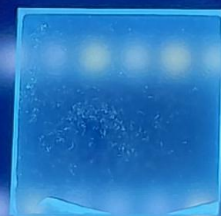
GS



GS



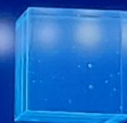
GS



GS



GS



GS



GS



GS



GS



GS



GS



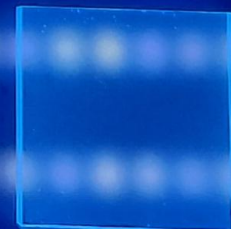
GS



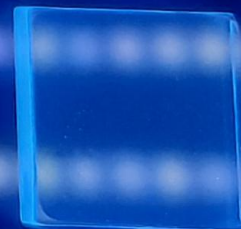
GS



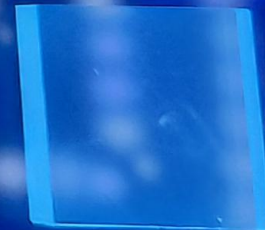
GS



GS



GS

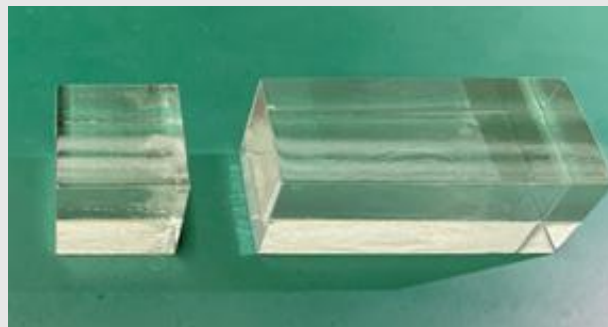


GS

0.2 What is the Glass Scintillator?



Plastic Scintillator



Glass Scintillator



Crystal Scintillator

Large density ★

High light yield ★★

Energy resolution ★

Low cost ★★ ★

Fast decay ★★ ★

Large size ★★ ★

Large density ★★

High light yield ★★

Energy resolution ★★

Low cost ★★

Fast decay ★★

Large size ★★

Large density ★★ ★

High light yield ★★ ★

Energy resolution ★★ ★

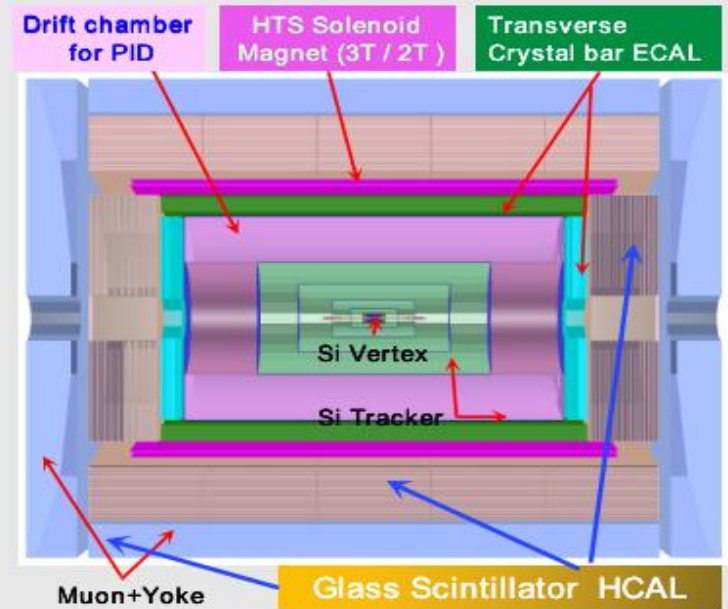
Low cost ★

Fast decay ★★

Large size ★★

Outline

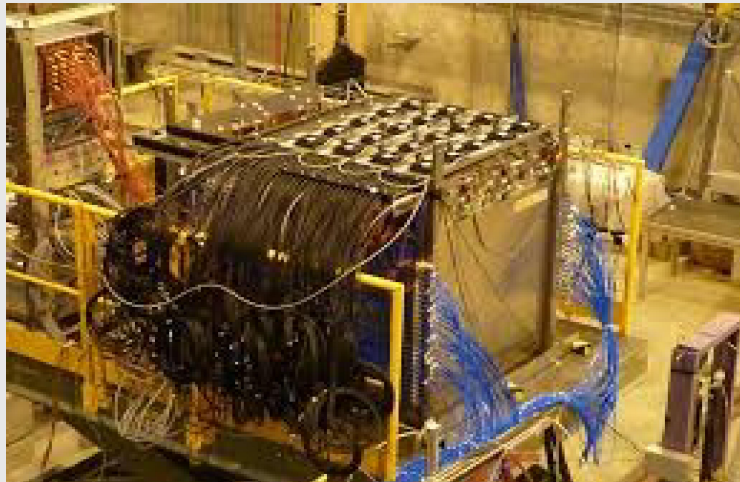
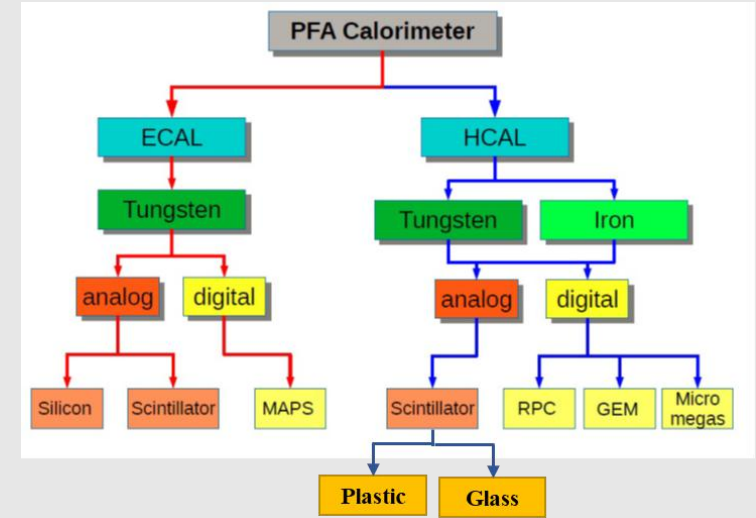
- 1. Motivation for HCAL
- 2. R&D of the GS
- 3. Performance of the GS
- 4. Summary and Next Plan



1.0 HCAL Design Options (The Past)

❑ Several HCAL design options have been proposed

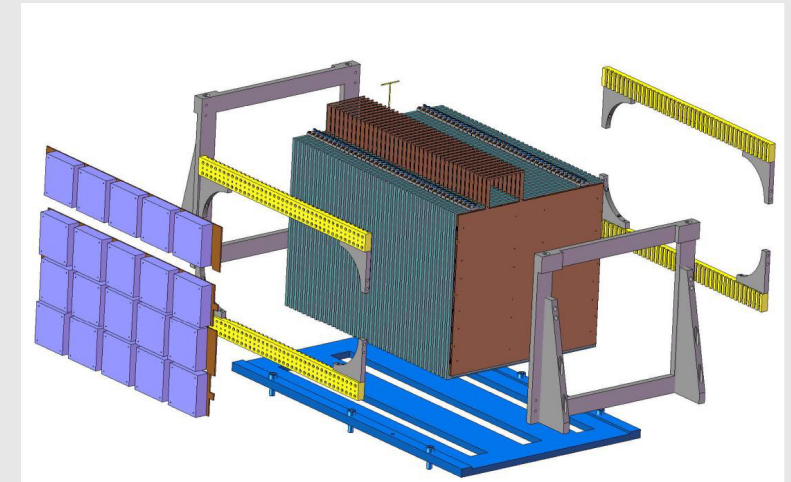
- Based on **Gaseous Detector**
 - e.g. CALICE SDHCAL [doi:10.1088/1748-0221/11/04/P04001](https://doi.org/10.1088/1748-0221/11/04/P04001)
- Based on **Liquid Argon**
 - e.g. ATLAS LAr Endcap HCAL [doi:10.1016/j.nuclphysbps.2011.03.150](https://doi.org/10.1016/j.nuclphysbps.2011.03.150)
- AHCAL: **Plastic Scintillator** & SiPM readout
 - e.g. CEPC AHCAL [doi:10.1088/1748-0221/17/11/P11034](https://doi.org/10.1088/1748-0221/17/11/P11034)



➤ CALICE SDHCAL Prototype

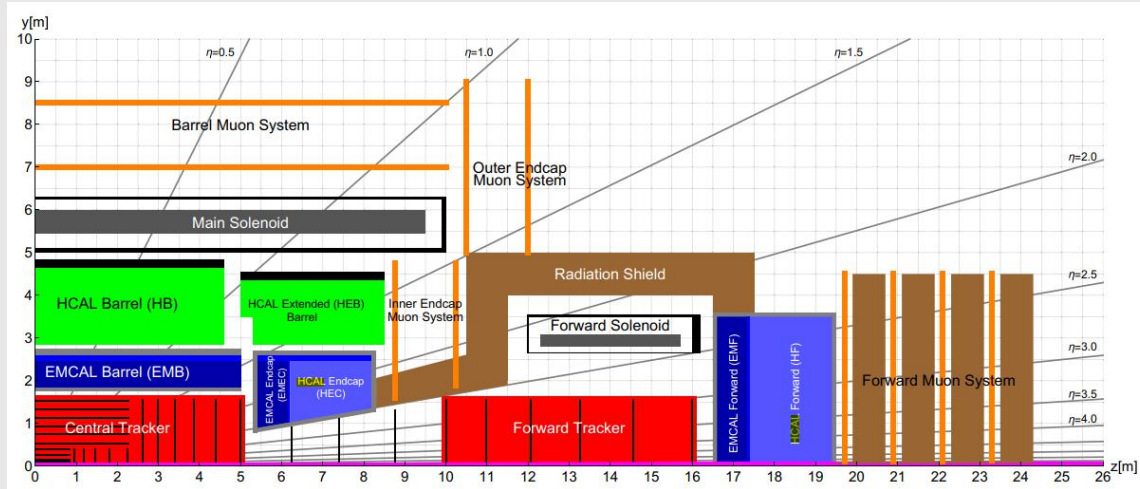


➤ ATLAS LAr Endcap HCAL



➤ CEPC AHCAL Prototype

1.1 HCAL Design Options (The Future)

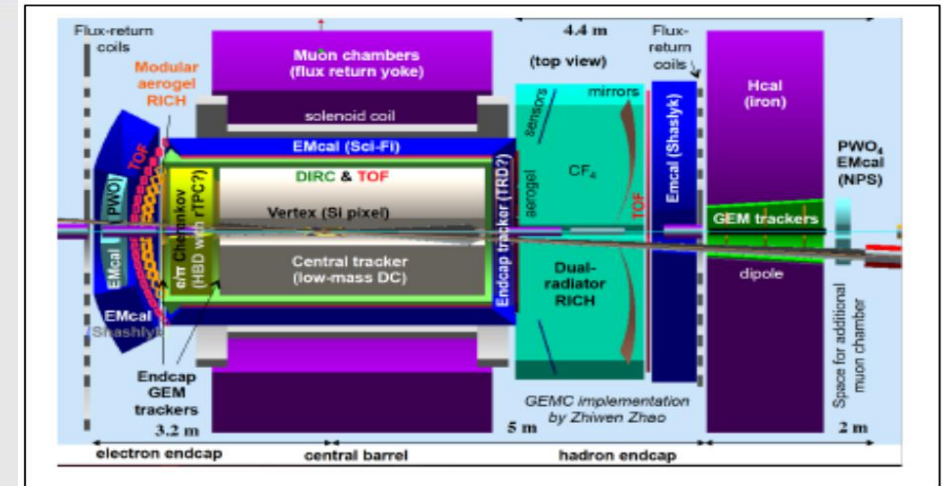


■ Longitudinal cross-section of the FCC-hh reference detector

Task 1.2: Hadronic section with optical tiles	
Subtask 1.2.1: AHCAL	Scintillating plastic tiles/Steel
Subtask 1.2.2: ScintGlassHCAL	Heavy glass tiles/Steel

Task 1.2

- AHCAL: Concept for continuous readout
- ScintGlass HCAL: cm scale tiles



■ Example layout of the EIC detector base design

Glass for EIC:

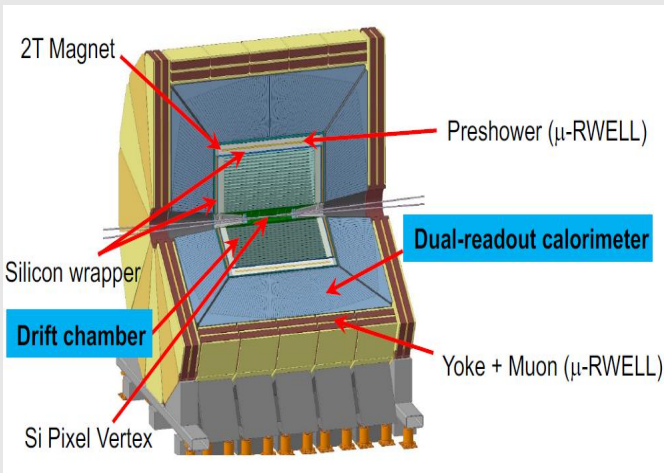
- Sufficient resolution
- Fast response
- Radiation hardness

Task 1 : build a cost effective glass-ceramic calorimeter for the outer endcaps and central regions

Task 2: Push the performance of **glass ceramics** towards crystals

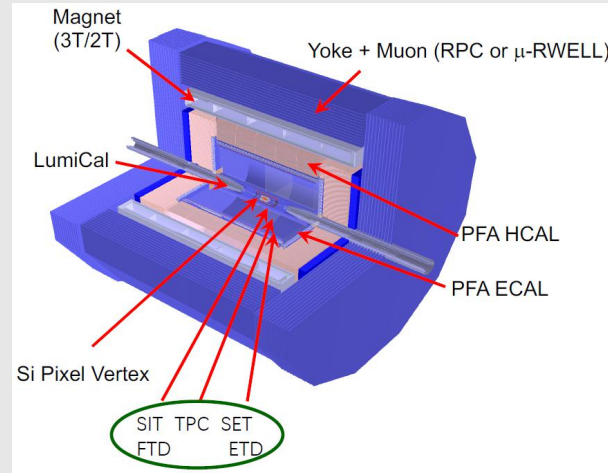
1.2 CEPC Conceptual Detector Design (The Future)

1st IDEA Concept (also proposed for FCC-ee)



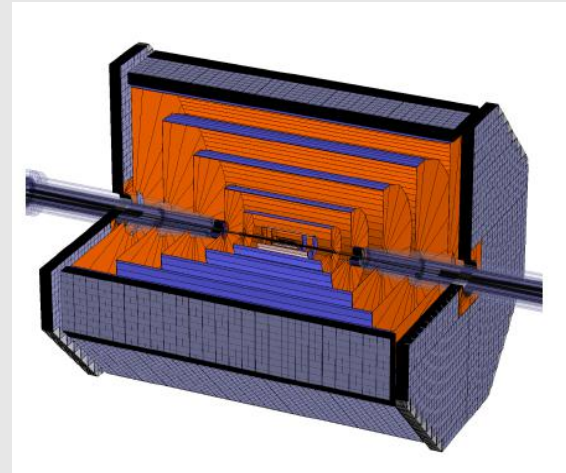
- Dual-readout calorimeter (Cerenkov-Fiber & Scint-Fiber)
- both for EM and Hadronic Shower

2nd CDR Baseline Design (Particle Flow Approach)



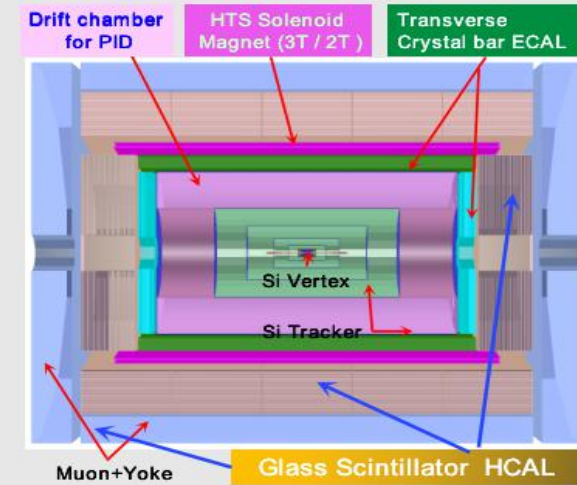
- AHCAL (PS/Steel) or SDHCAL (Gas/Steel)
- Si/W ECAL or PS/W ECAL

3rd FST concept (Full Silicon Tracker)



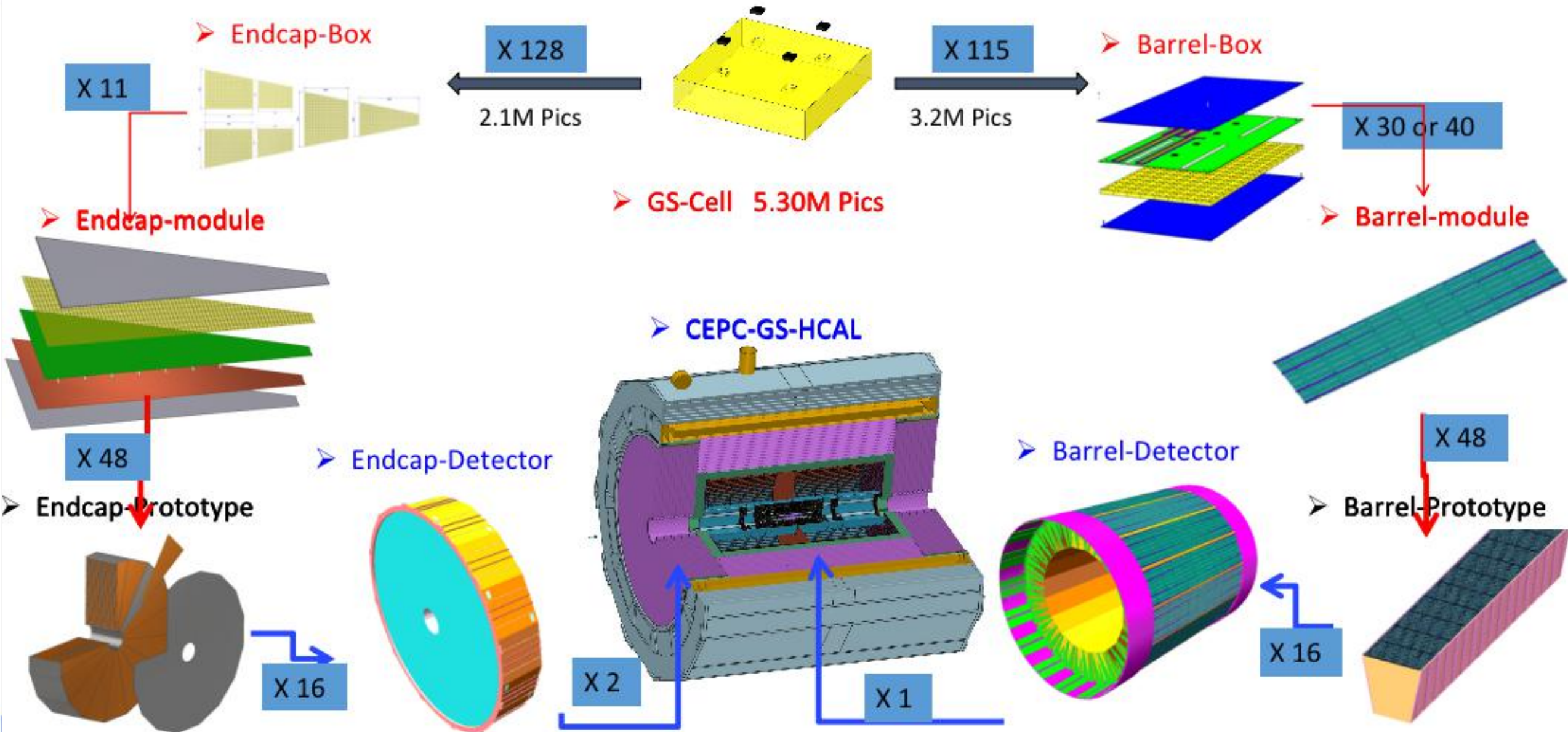
- AHCAL (PS/Steel) or SDHCAL (Gas/Steel)
- Si/W ECAL or PS/W ECAL

4th Conceptual Detector



- GSHCAL (GS/Steel)
- Crystal ECAL

The GS-HCAL for CEPC (The Future)



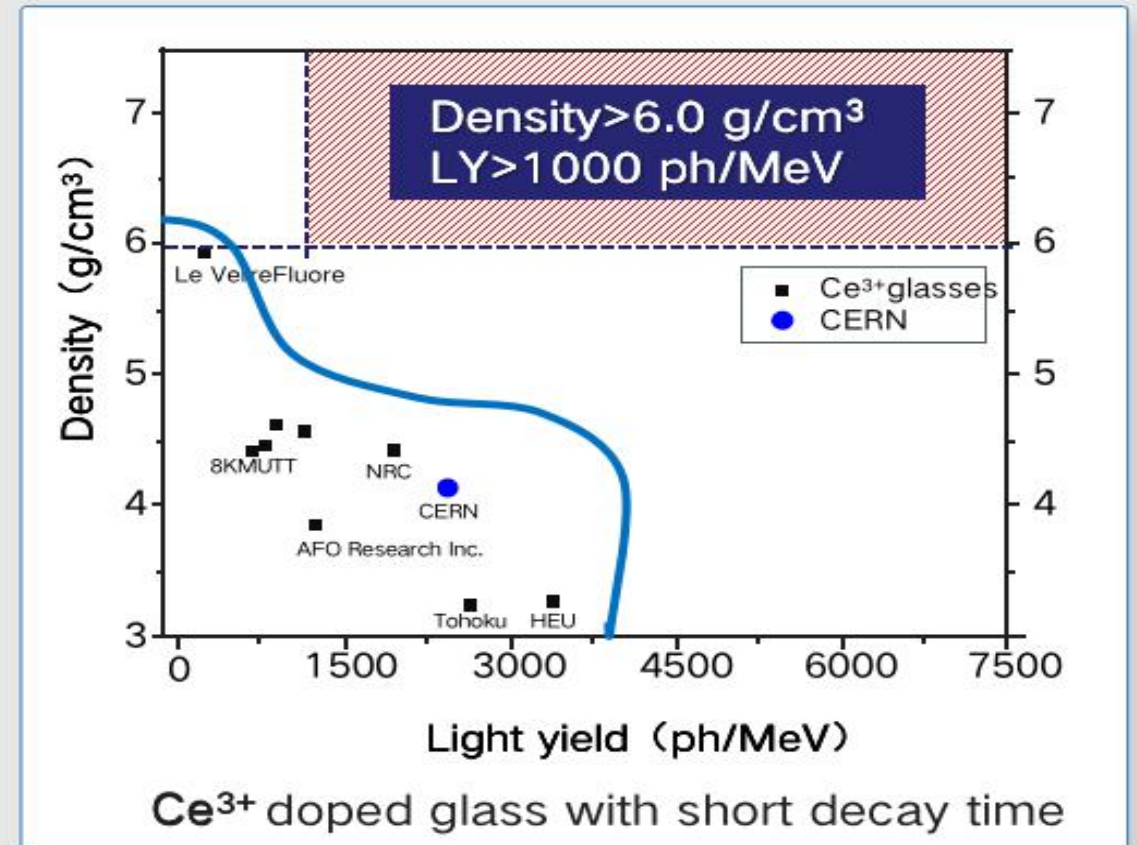
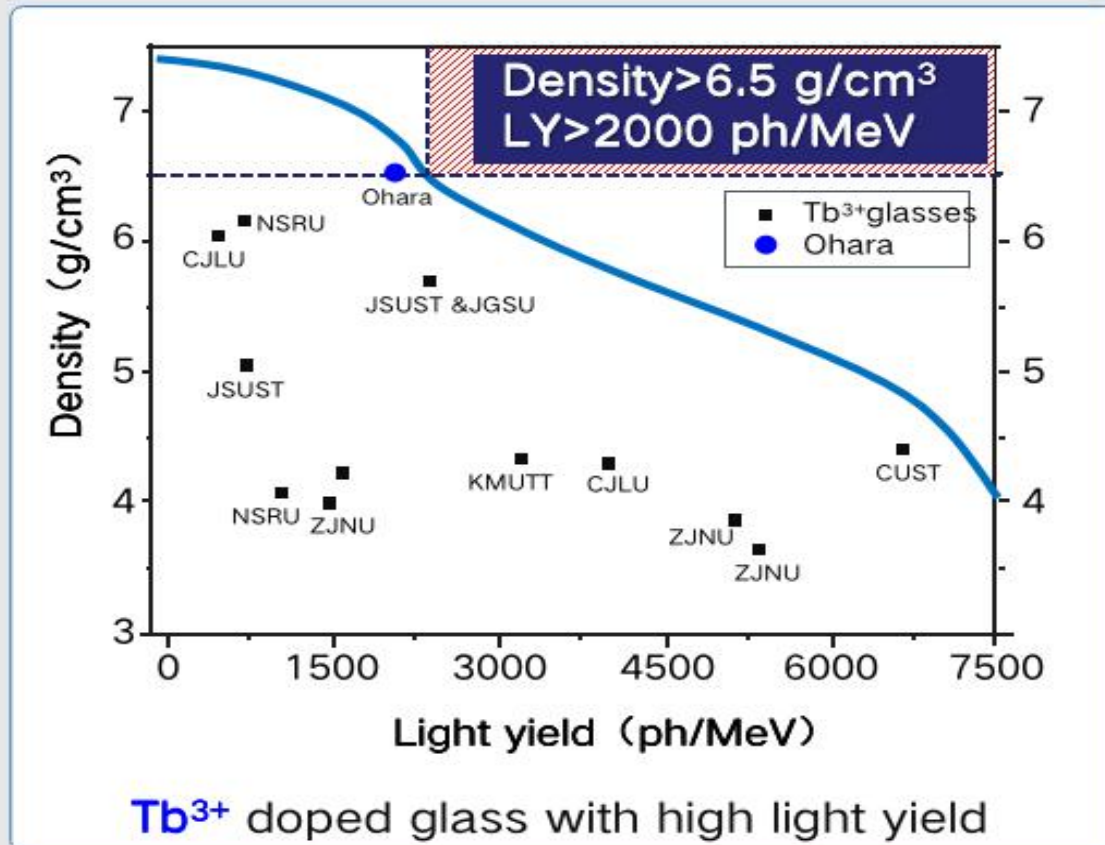
Outline

- 1. Motivation for HCAL
- 2. R&D of the GS
- 3. Performance of the GS
- 4. Summary and Next Plan

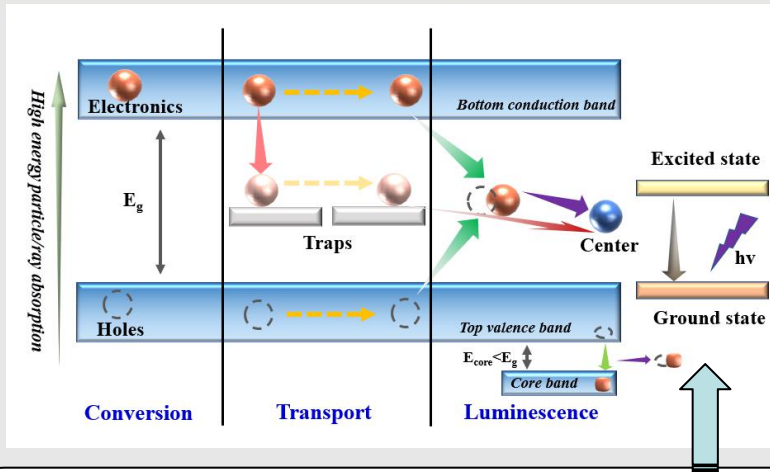


2.1 Current Research Status of the GS

- Before 2000, the high-density GS is mainly based on Pb (plumbum) or Bi (bismuth), with poor scintillation light;
- After 2000, GS with lanthanide elements (Tb, Terbium; Ce, Cerium) attract more attention for improved LY
- However, it's a great challenge to realize a **high density** and **high light** yield and **Low cost** at the same time.



2.2 The Design of the GS



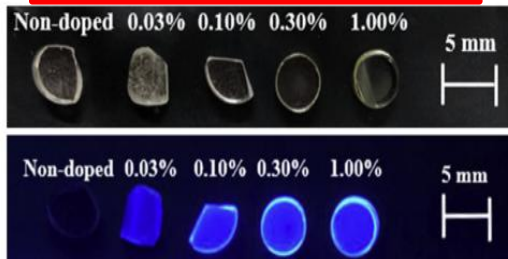
➤ **Scintillation mechanism----** **Luminescence Center**

➤ **Conversion**—photoelectric effect and Compton scattering effect;

➤ **Transport**—electrons and holes migrate;

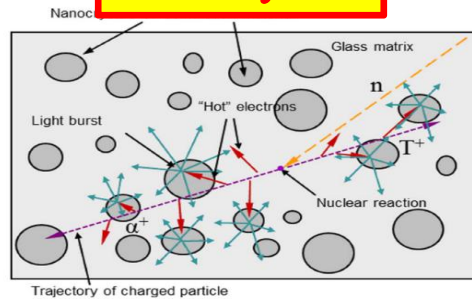
➤ **Luminescence**—captured by the luminescent center ions

Lanthanide elements



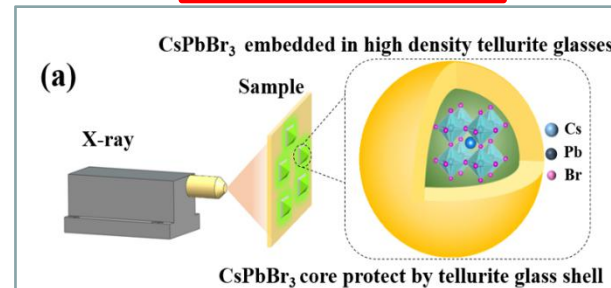
Journal of Alloys and Compounds
782 (2019) 859-864

Nanocrystals



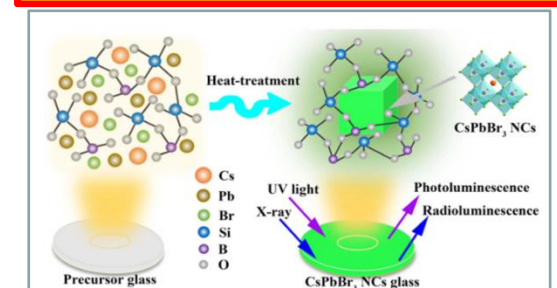
IEEE TNS 60 (2) 2013

Quantum Dots



Optics Letters 46(14) 3448-3451 (2021)

Lanthanide + Quantum Dots



Vol. 9, No. 12 / 2021 / Photonics Research

■ High Light Yield (> 2000 ph/MeV): Lanthanide for the Luminescence Center: **Cerium (Ce)**;

■ High Density (> 6 g/cm³) and Low radioactivity background: **Gadolinium (Gd)**; ~~lutetium (Lu)~~

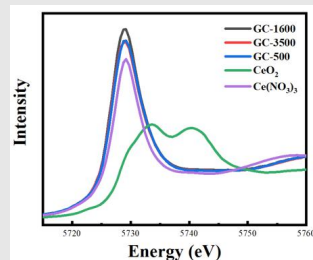
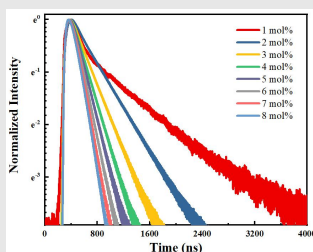
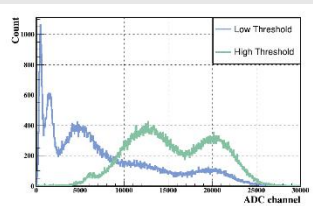
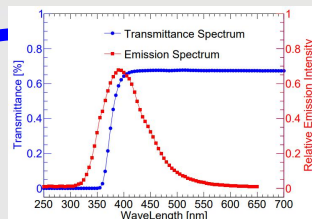
2.3 Large Area Glass Scintillator Collaboration



Spokesperson: Sen QIAN

- The Glass Scintillator Collaboration Group established in Oct.2021;
- There are 4 Institutes from CAS, 6 Universitys, 3 Companies join us for the R&D of GS;

2.4 The Scintillator Test Facilities for GS



Others

.....

- Transmittance
- Absorbance
- Refractive index
- Emission peak
- Light yield
- Energy resolution
- MIP response
- Neutron discrimination
- Rise time
- Fall time
- Decay time
- Afterglow
- Coincidence time
- Valence state
- Coordination
- Elemental analysis
- Structural analysis
- Zeeman effect
- Radiation resistance
- Homogeneity

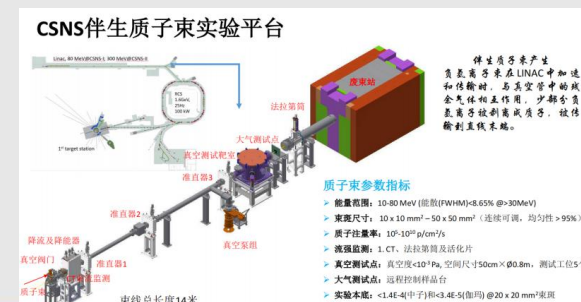
➤ IHEP--PMT Lab for Scintillator Test



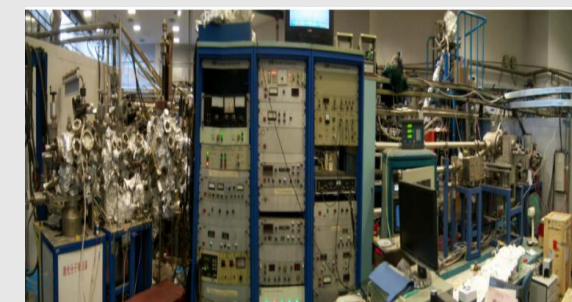
➤ IHEP--Radioactive Test



➤ IHEP-CSN-- P Beam



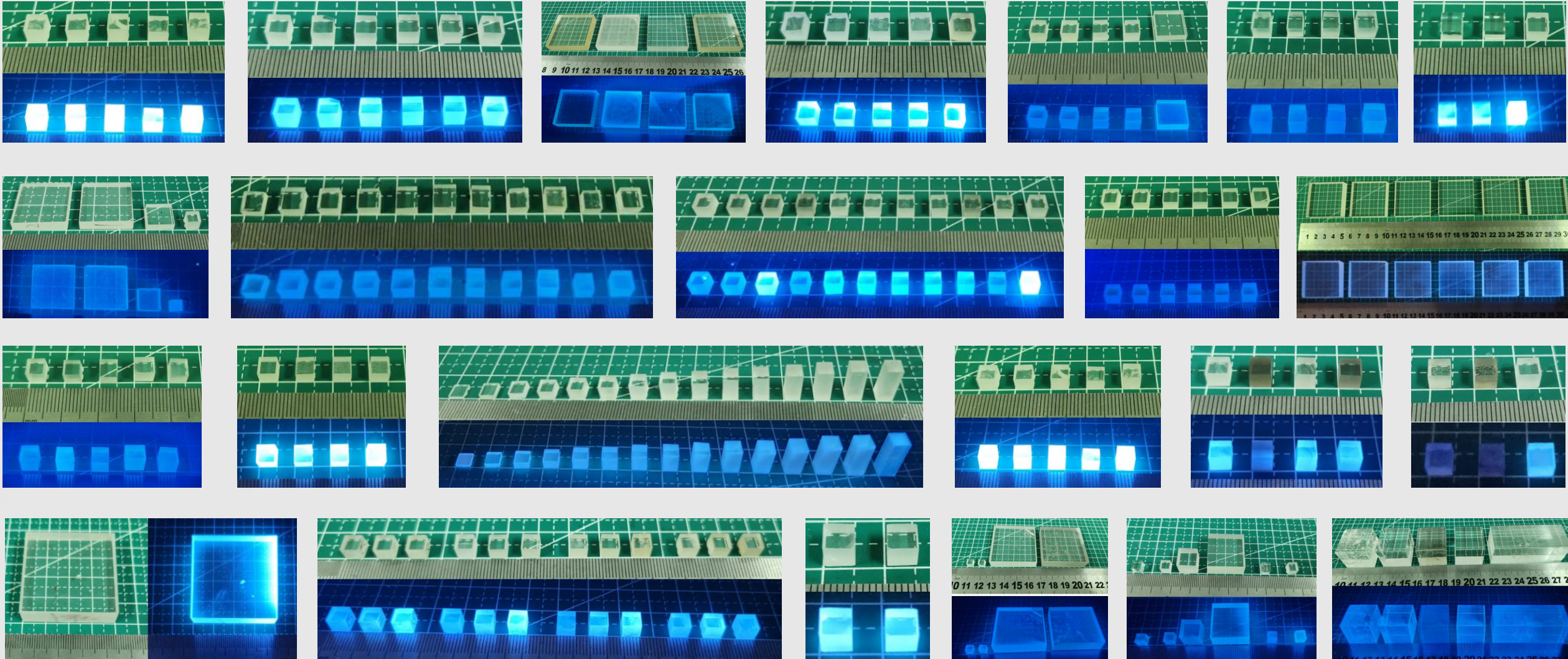
➤ IHEP--XAFS



➤ CERN-MUON Beam



2.5 The GS Samples produced (>1000)



Outline

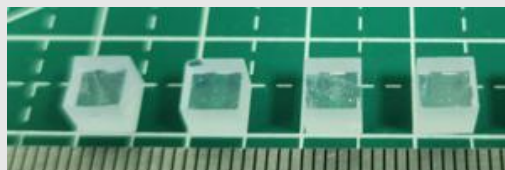
- 1. Motivation for HCAL
- 2. R&D of the GS
- 3. Performance of the GS
- 4. Summary and Next Plan



3.1 The Light Yield and Decay Time

GFO & GFO+ (5mm × 5mm × 5mm)

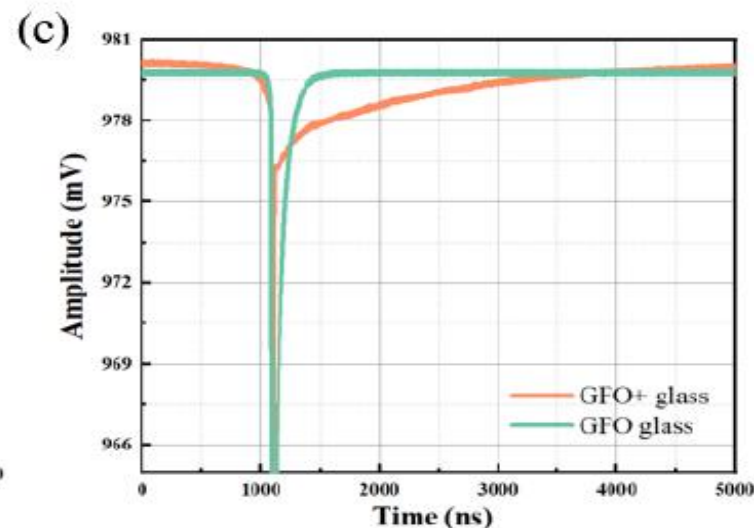
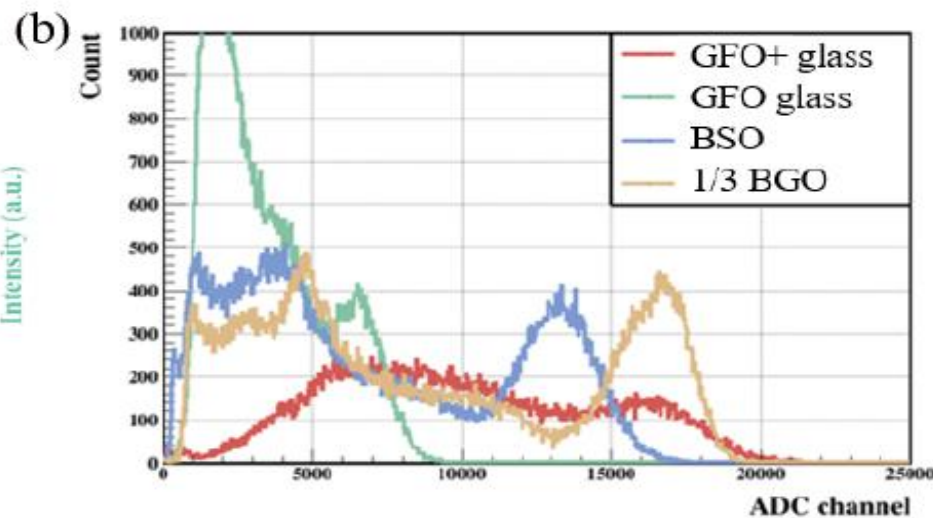
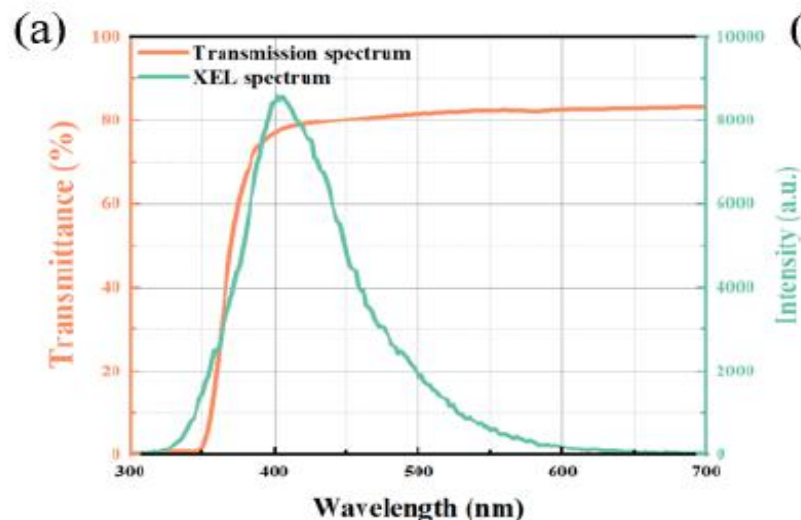
GFO



- Density~6.0 g/cm³
- LY=985 ph/MeV
- ER=30.3%
- LO in 1μs=982 (99%)
- Decay=36 (8%), 105 ns

- Density~6.0 g/cm³
- LY=2445 ph/MeV
- ER=25.8%
- LO in 1μs=1074 (44%)
- Decay=101 (2%), 1456 ns

GFO+



➤ (a) Transmission and XEL spectra

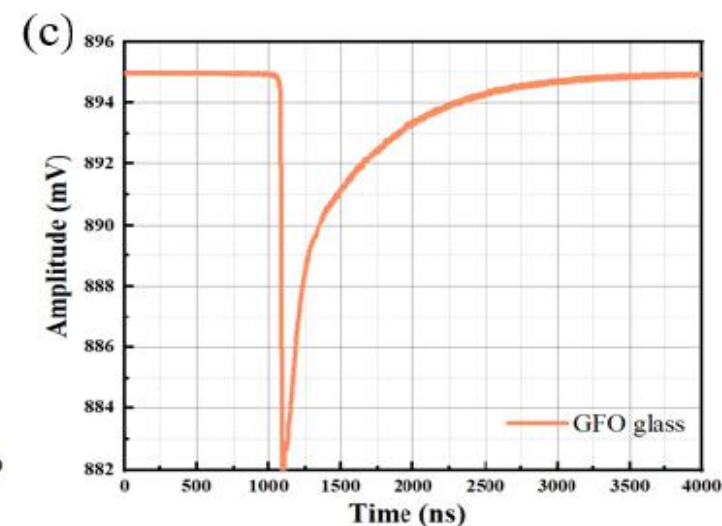
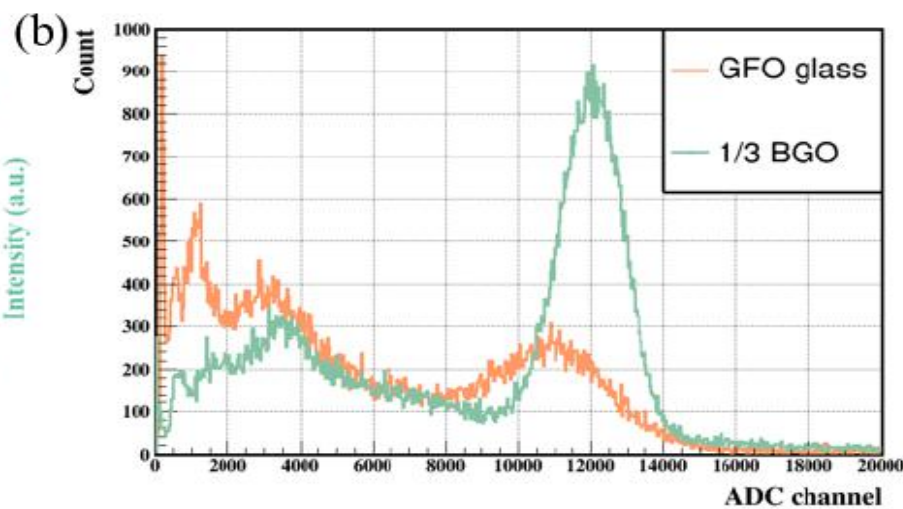
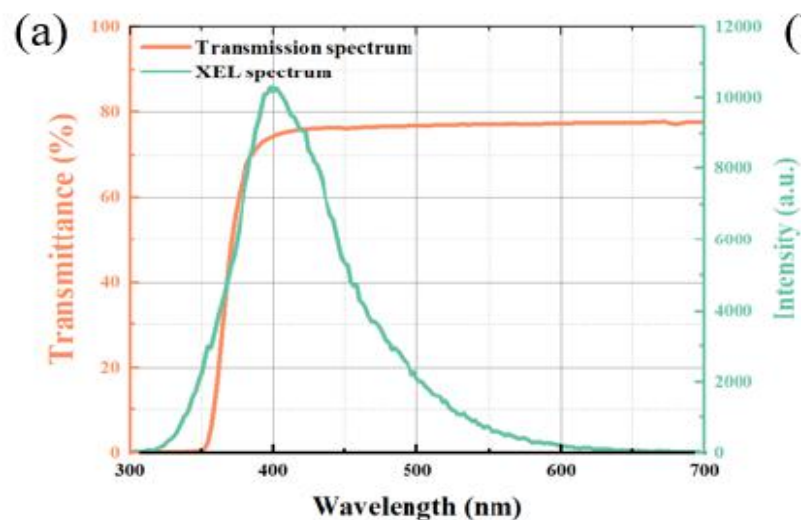
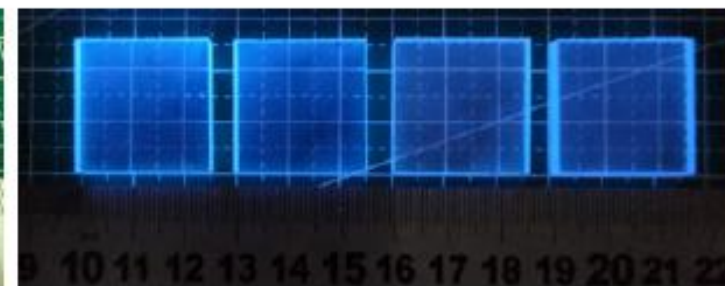
➤ (b) Energy spectra

➤ (c) Scintillation decay curves

3.1 The Light Yield and Decay Time

GFO (40mm × 40mm × 10mm)

- Density=6.0 g/cm³
- LY=1506 ph/MeV
- ER=41.2%
- LO in 1μs=1129 (75%)
- Decay=53 (3%), 655 ns

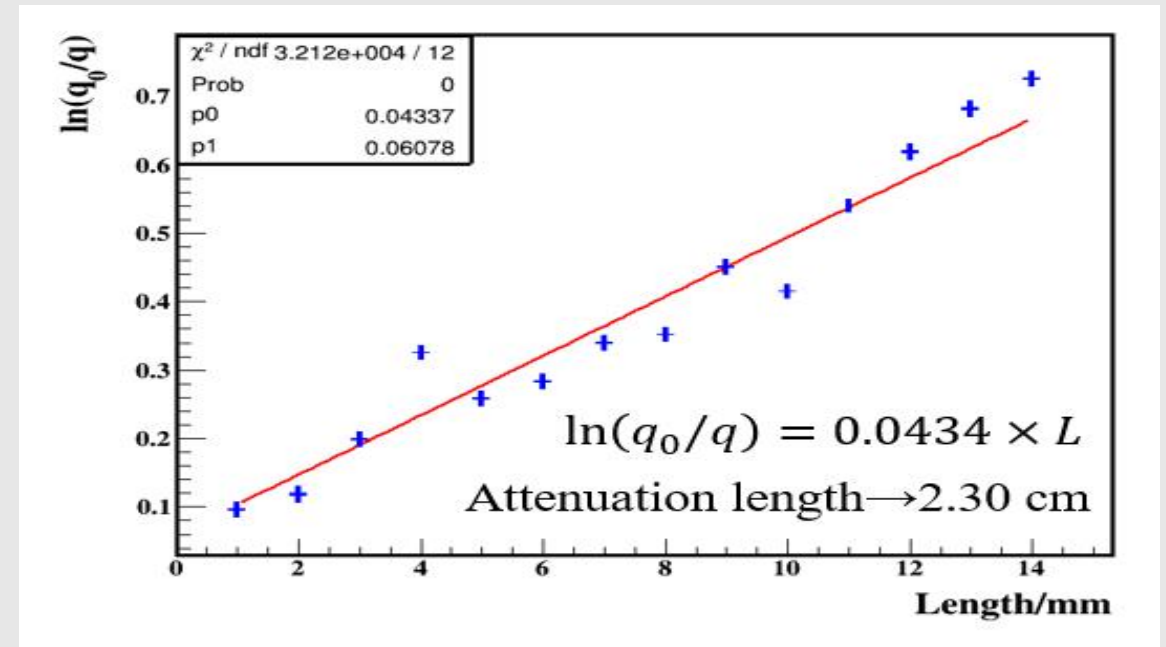
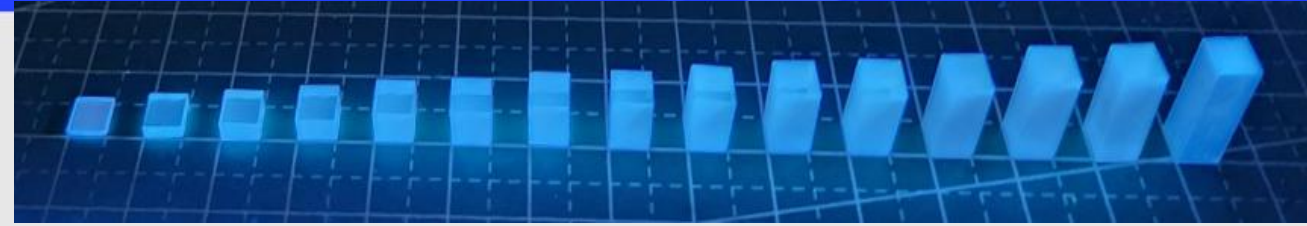
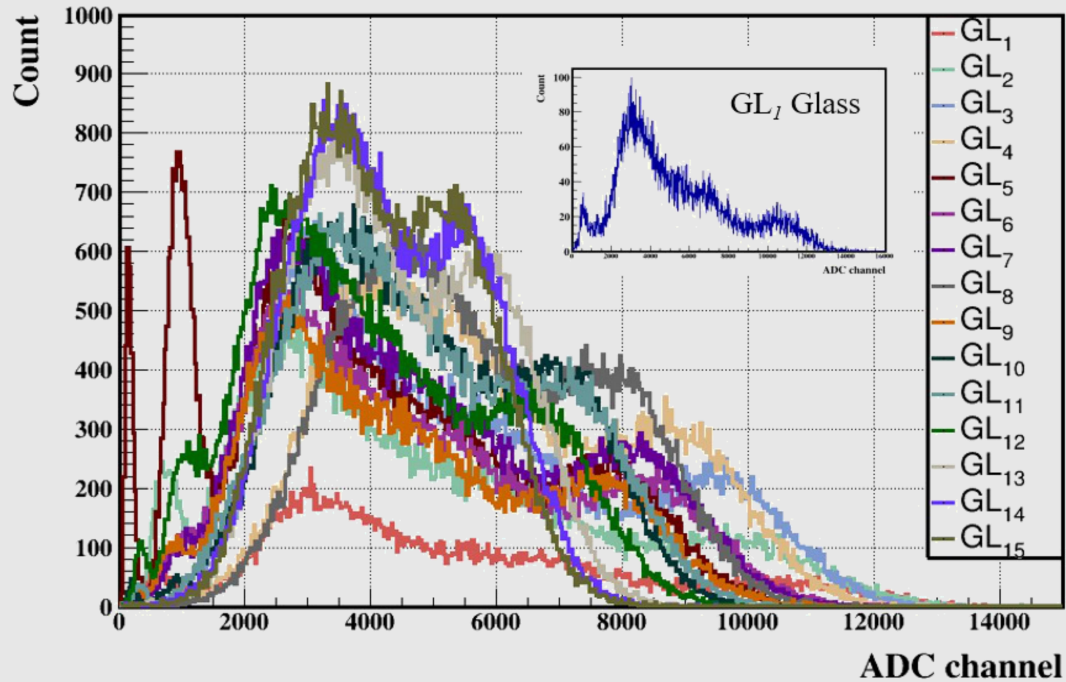


➤ (a) Transmission and XEL spectra

➤ (b) Energy spectra

➤ (c) Scintillation decay curves

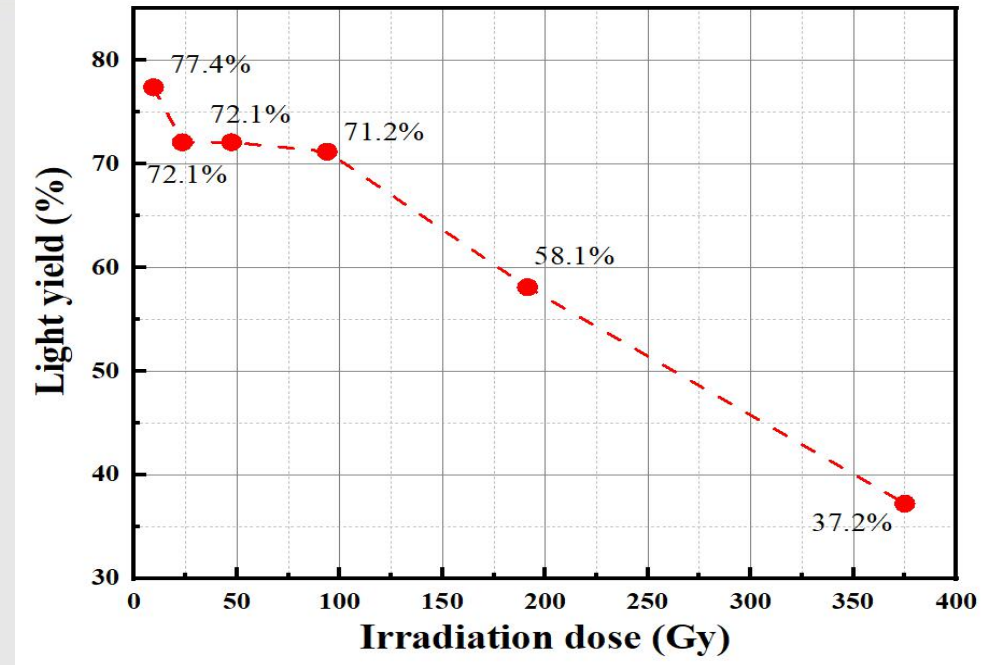
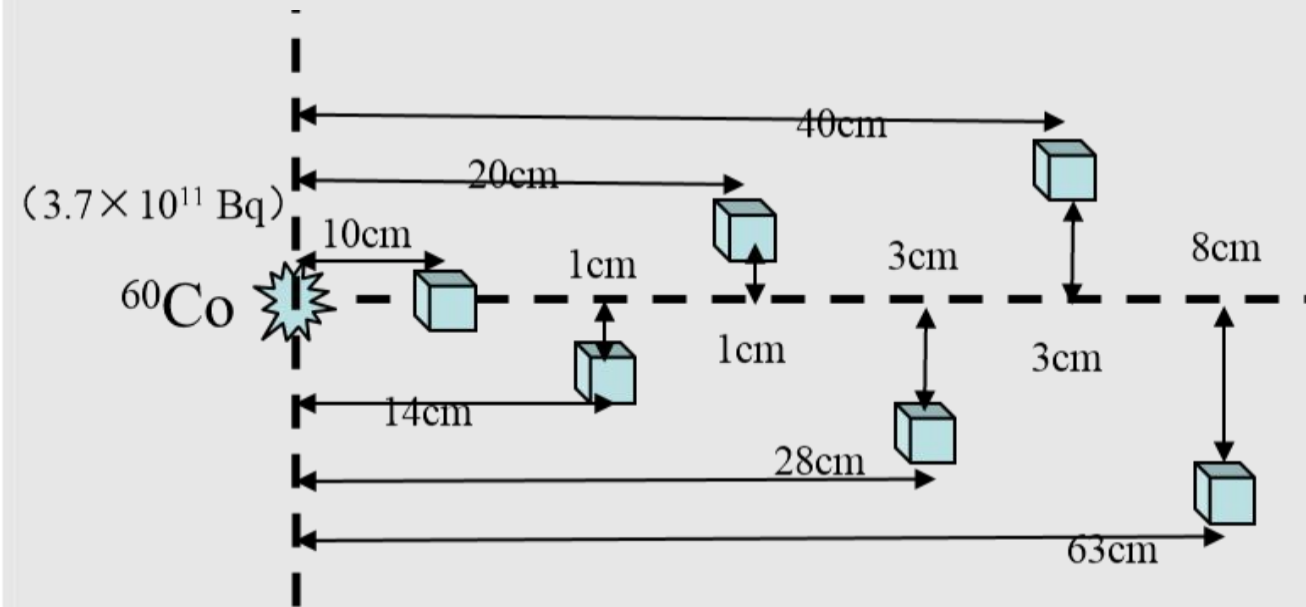
3.2 Actual Attenuation length Test



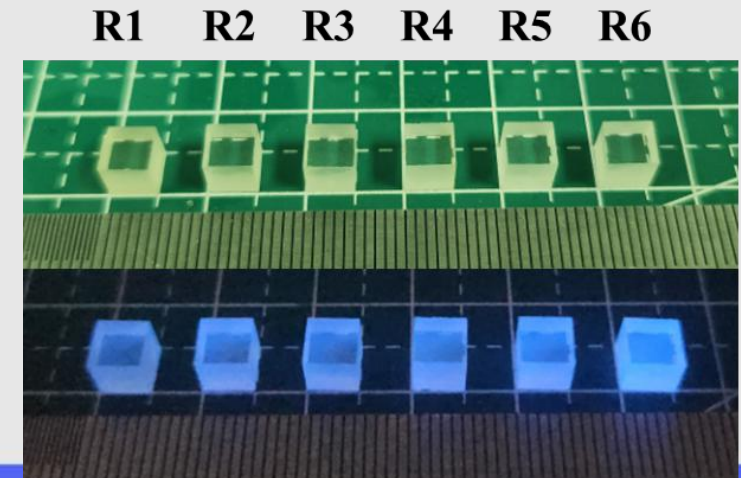
■ GS of 1000 ph/MeV grade:

- actual attenuation length $\rightarrow 2.30 \text{ cm}$ (around emission peak 400nm) $\ll 3.01\text{-}6.46 \text{ cm}$
- Defects and self-absorption in glass $\rightarrow$ lower light attenuation length

3.3 Irradiation Test of GS with ^{60}Co

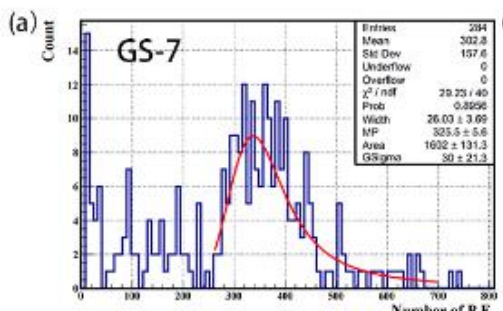
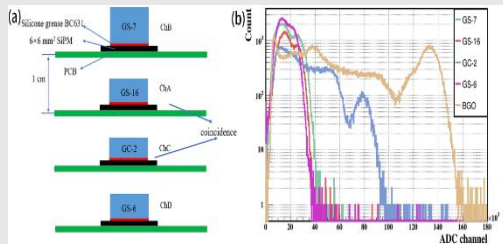


- 6 glass samples ($5 \times 5 \times 5 \text{ mm}^3$) were irradiated with a ^{60}Co source of $3.656 \times 10^{11} \text{ Bq}$ and were placed with different distance to obtain different dose
- The light output of the glass sample will be reduced to **58%** of its original level after exposure to a dose of $\sim 200 \text{ Gy}$.



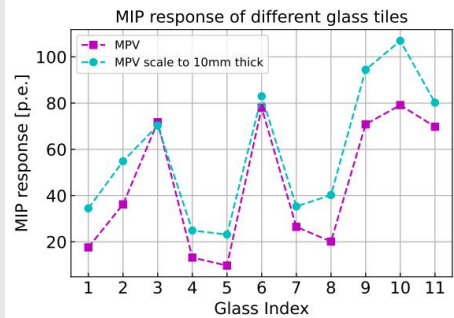
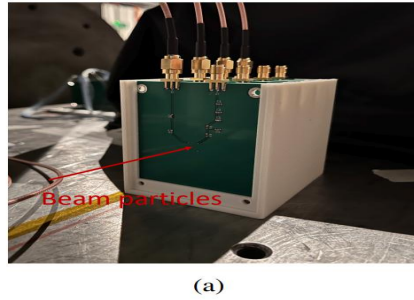
3.4 MIP Response of the GS

1st Cosmic ray (Integral gate=1 μ s)



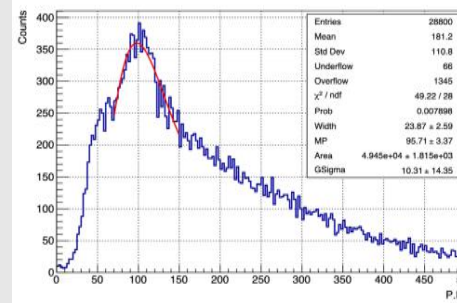
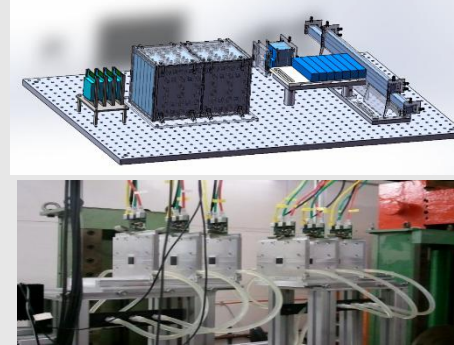
- Lab grade GS (5mm cube)
- Normalize the thickness and density of GS
- Prove consistency of LY and MIP responses

CERN Muon beam (Integral gate=1 μ s)



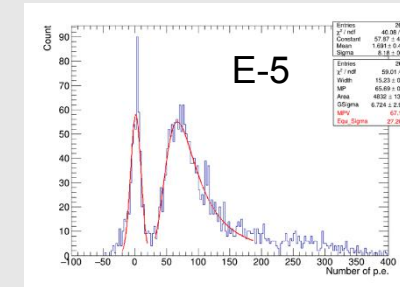
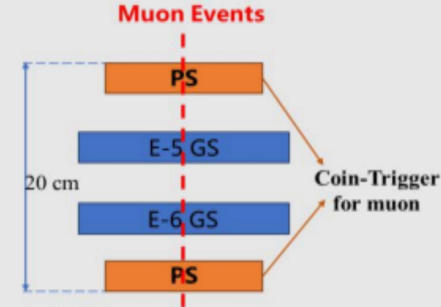
- $(25-40) \times (25-35) \times (4-10) \text{ mm}^3$
- Typical light yield: 500-600 ph/MeV
- Typical MIP response: 60-100 P.E./MIP

DESY Electron beam (Integral gate=1 μ s)



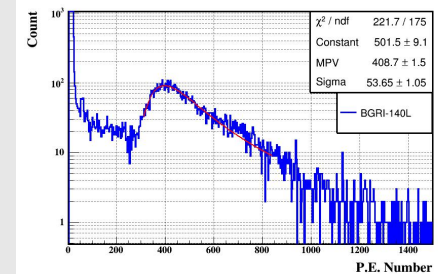
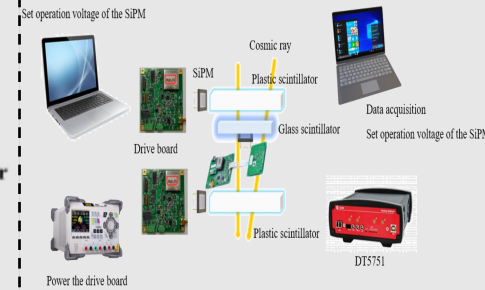
- Size=40*40*10 mm³
- Typical light yield: 600-700 ph/MeV
- Typical MIP response: 80-90 P.E./MIP

2nd Cosmic ray (Integral gate=4 μ s)



- Size=40*40*10 mm³
- Light yield: ~750 ph/MeV
- MIP response: ~135 P.E./MIP

3rd Cosmic ray (Integral gate=4 μ s)



- Size=20*20*10 mm³
- Light yield: ~1800 ph/MeV
- MIP response: ~400 P.E./MIP

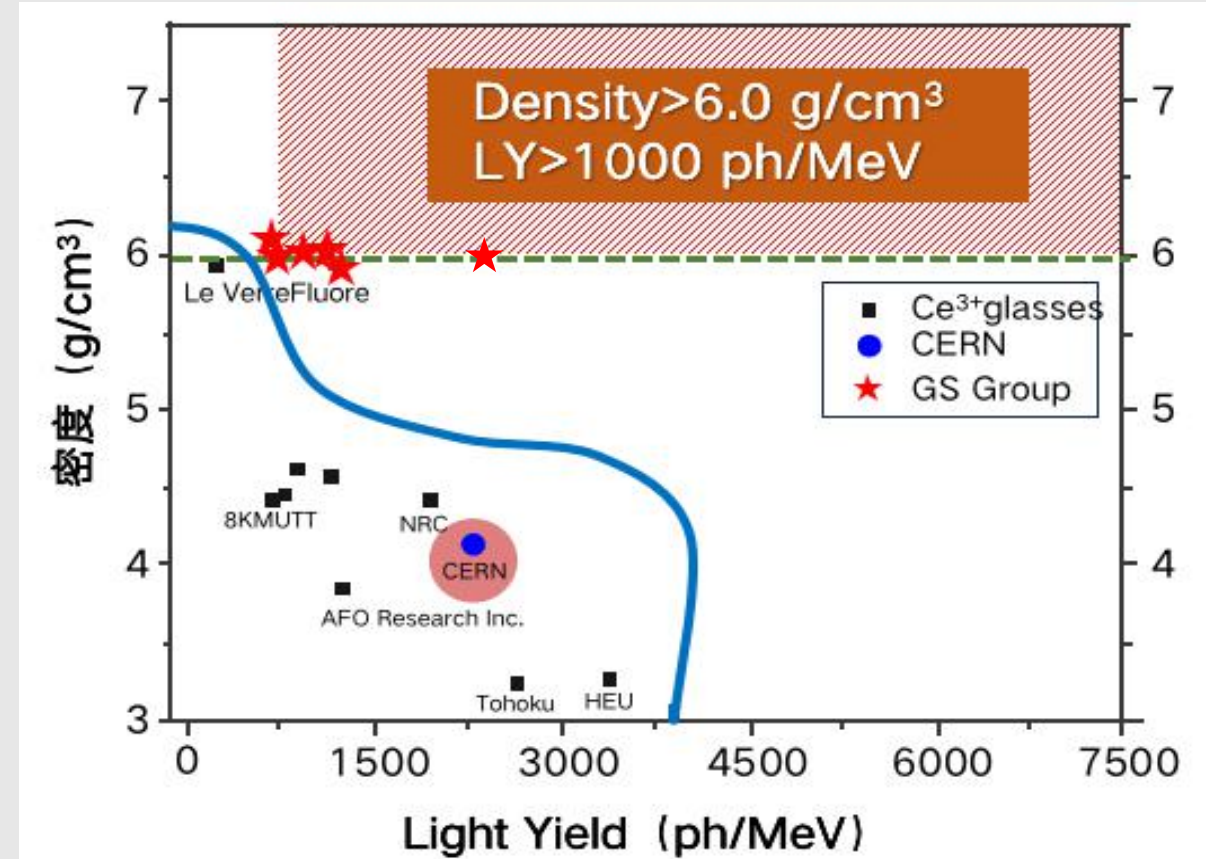
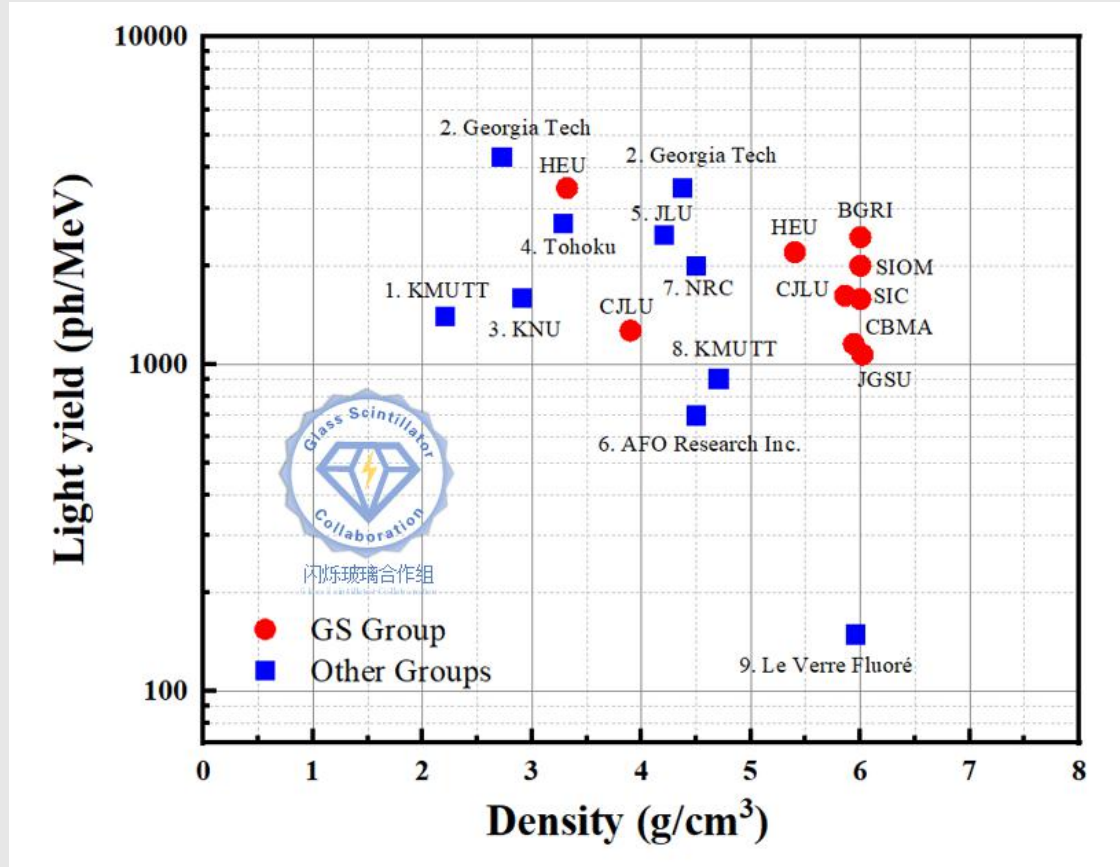
Outline

- 1. Motivation for HCAL
- 2. R&D of the GS
- **3. Performance of the GS**
- 4. Summary and Next Plan

4.0 Technical challenges--Glass Scintillator

Parameters	Unit	BGO	BSO	DSB Glass	GFO	GFO+
Density	g/cm ³	7.13	6.8	4.2	6.0	6.0
Melting point	°C	1050	1030	1550	1250	1150
Radiation Length, X ₀	cm	1.12	1.15	2.62	1.59	1.64
Molière radius	cm	2.23	2.33	3.33	2.49	2.50
Nuclear interaction length	cm	22.7	23.4	31.8	24.2	24.1
Z _{eff}	--	71.5	73.8	49.7	56.6	56.9
dE/dX	MeV/cm	8.99	8.59	5.9	8.0	8.0
Emission peak	nm	480	470	430	400	390
Refractive Index	--	2.15	2.68	-	1.74	1.76
Light yield, LY	ph/MeV	7500	1500	2500	985	2445
Energy resolution, ER	%662keV	9.5	32.0	-	30.3	25.8
Decay time	ns	60, 300	100	90, 400	36, 105	101, 1456
Cost	\$/cc	6.0	7.0	?	<1.0	<1.0

4.1 GS Group Samples

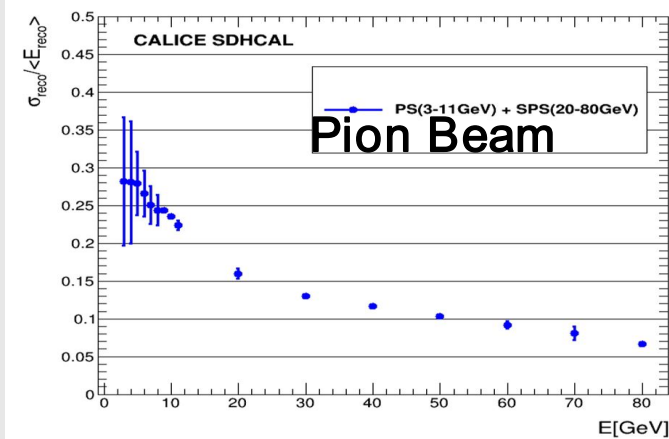
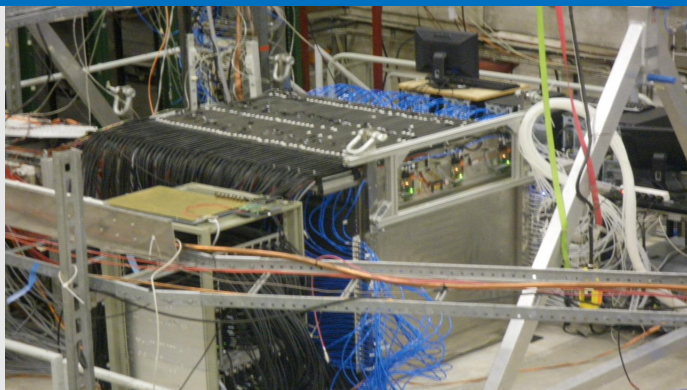


- The GS group has carried out a comprehensive and complete study;
- For high density glass scintillator, the light yield of GS group samples is in the absolute lead.

4.2 Next Plan

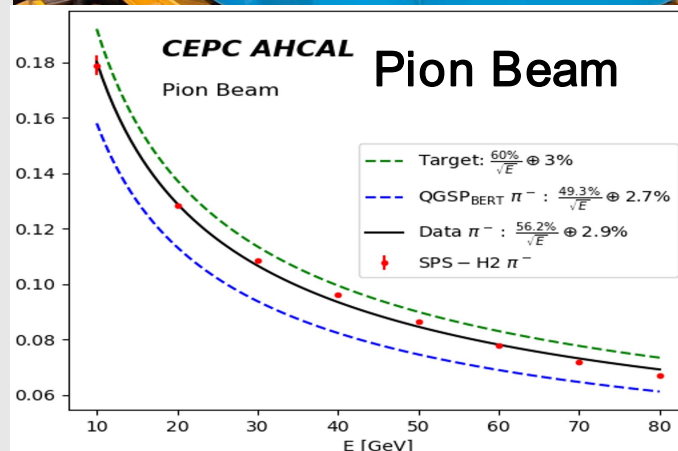
Comparison of Energy Resolution of HCAL Prototypes

RPC-SDHCAL, 48-layer, 1x1 cm²



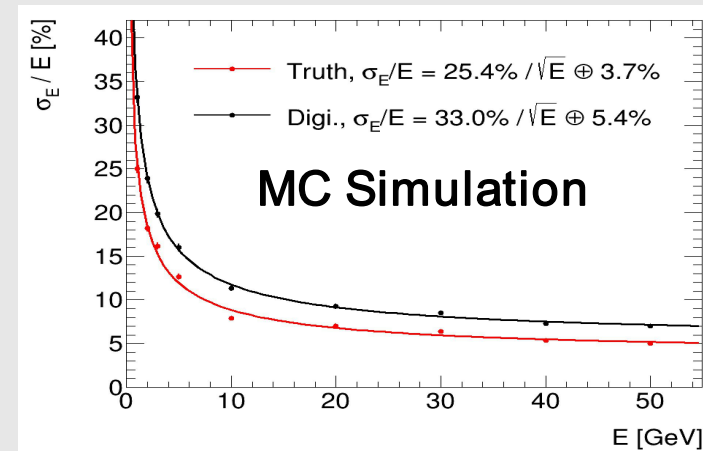
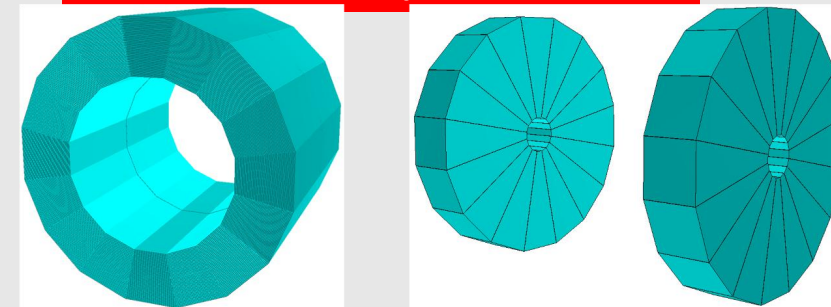
$$\sigma_E/E = \frac{65\%}{\sqrt{E}} \oplus 2.5\%$$

PS-HCAL, 40-layers, 4x4 cm²



$$\sigma_E/E = \frac{56.2\%}{\sqrt{E}} \oplus 2.9\%$$

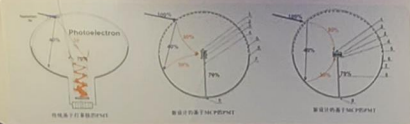
GS-HCAL, 48 layers, 4x4 cm²



$$\sigma_E/E = \frac{33.0\%}{\sqrt{E}} \oplus 5.4\%$$

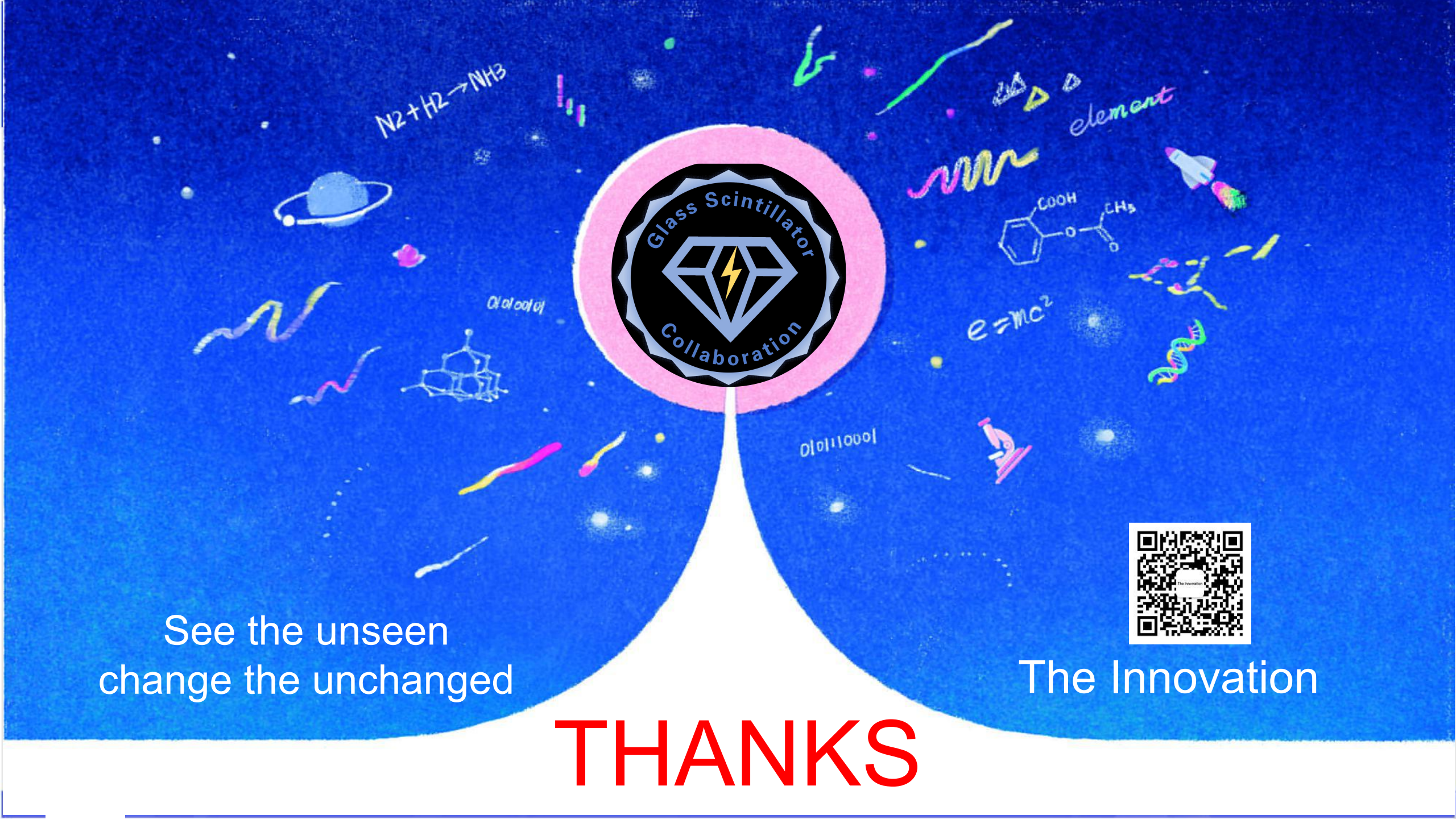
新型MCP-PMT研制

根据在研制过程中大量使用光电倍增管的经验，研制下一代大型中微子实验计划需要，结合国际上最新光电倍增管研制技术，设计、制造和测试一种国际上尚无同类产品、具有自主知识产权、与超导磁体结合的光电倍增管的技术方案。



知识产权保障：
专利名称：一种光电倍增管
专利号：ZL201010111111.1
专利权人：中国科学院高能物理研究所
发明人：王恩科、王恩科、王恩科
发明人：王恩科、王恩科、王恩科
发明人：王恩科、王恩科、王恩科
发明人：王恩科、王恩科、王恩科





See the unseen
change the unchanged

The Innovation

THANKS